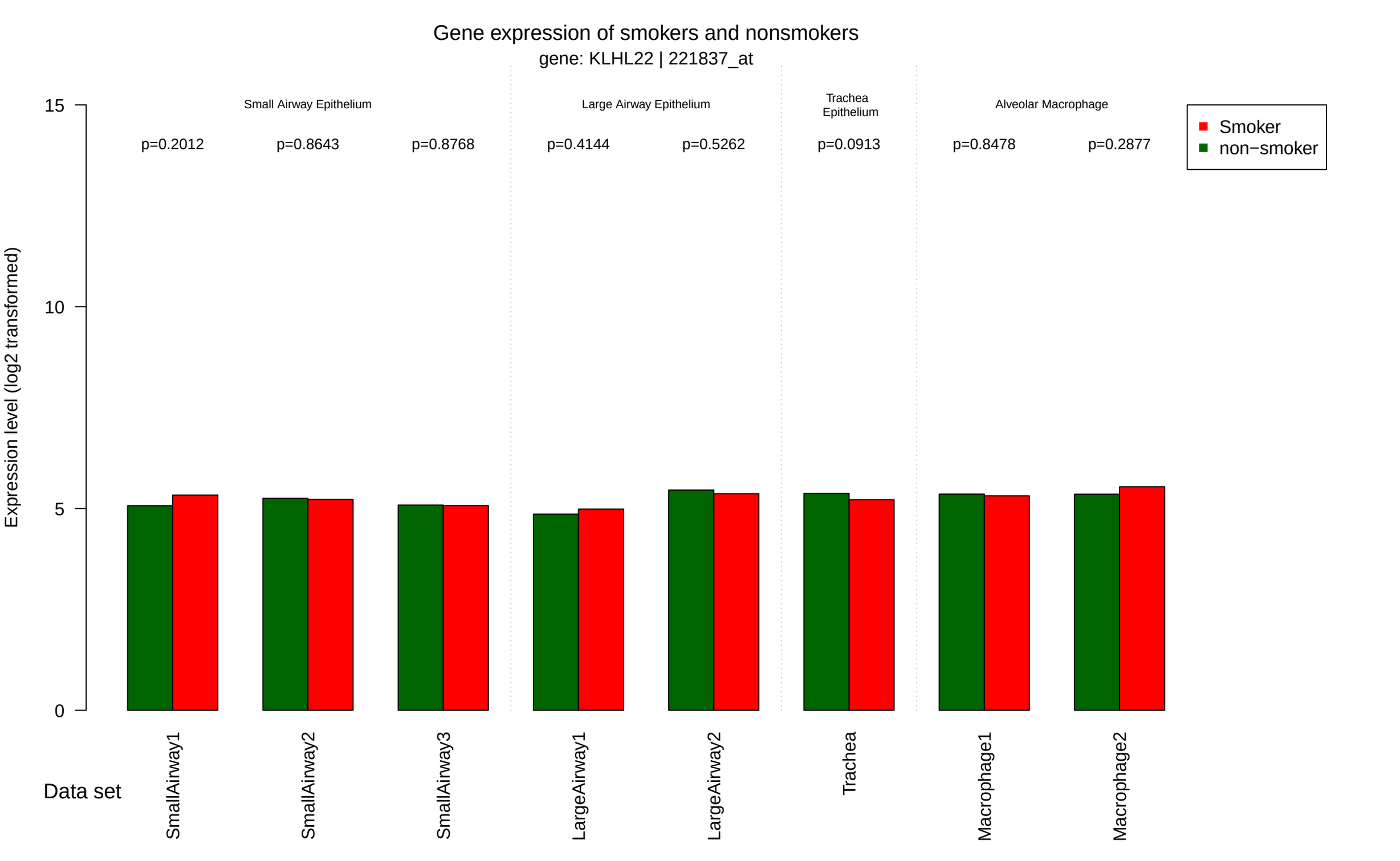
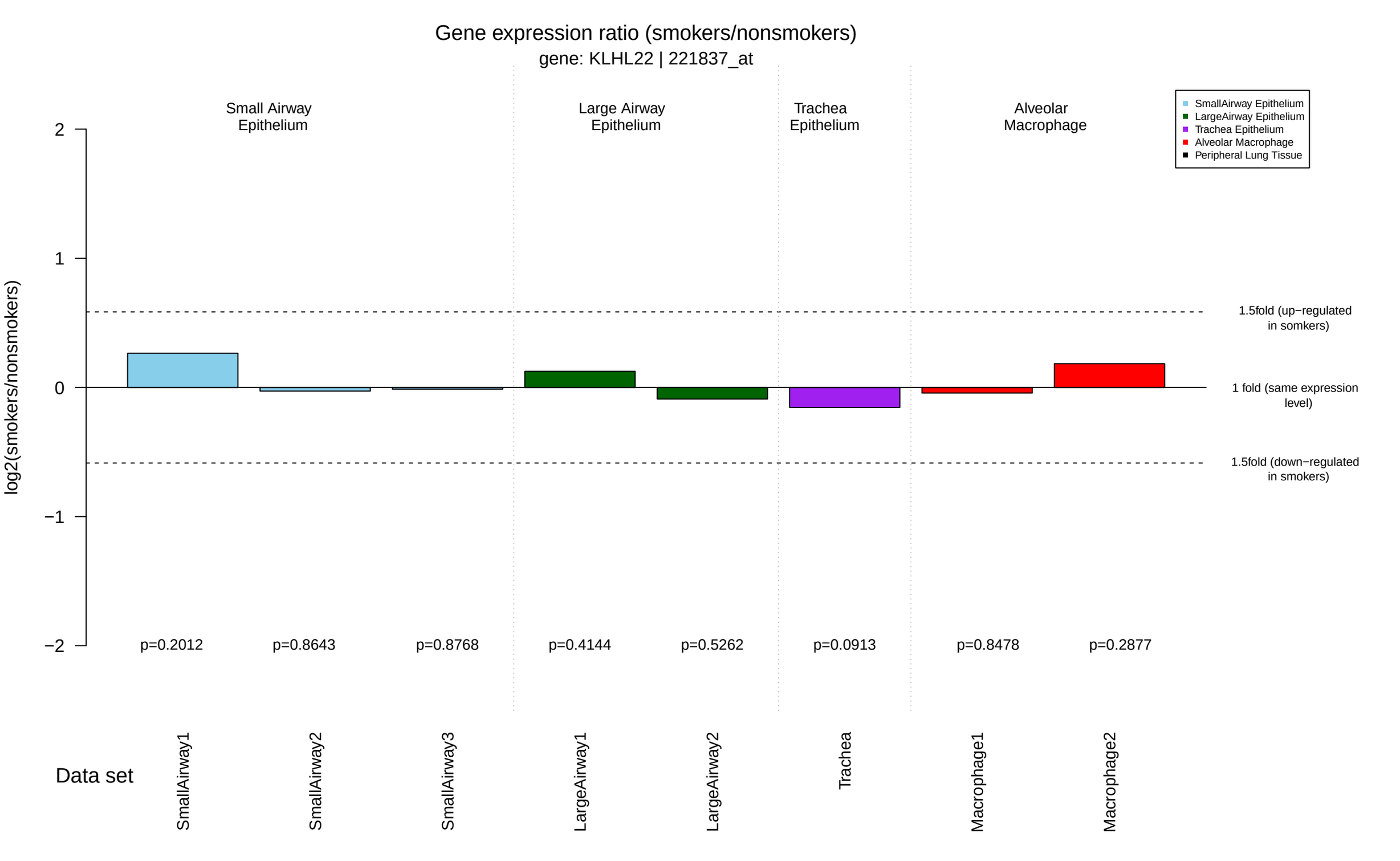
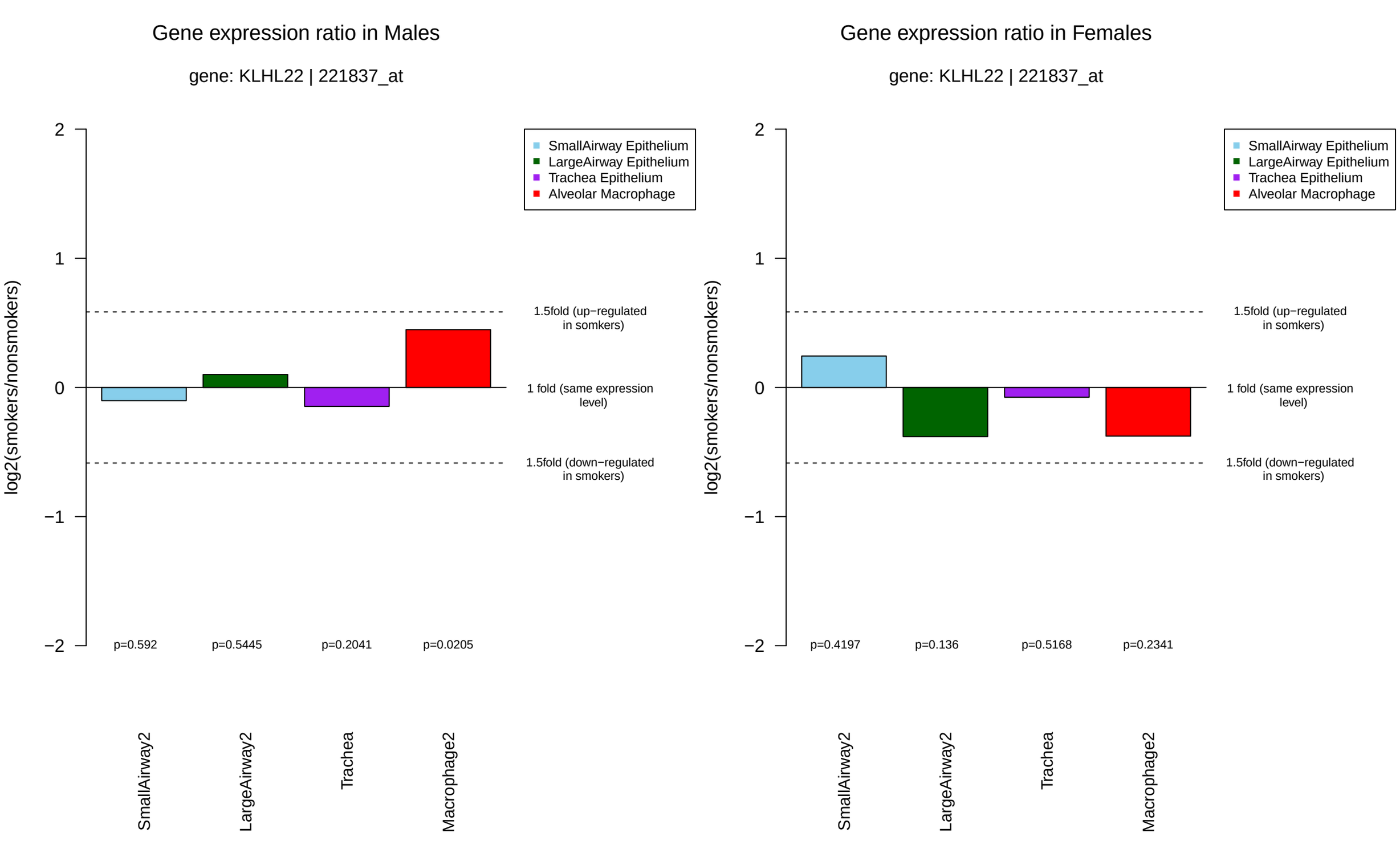
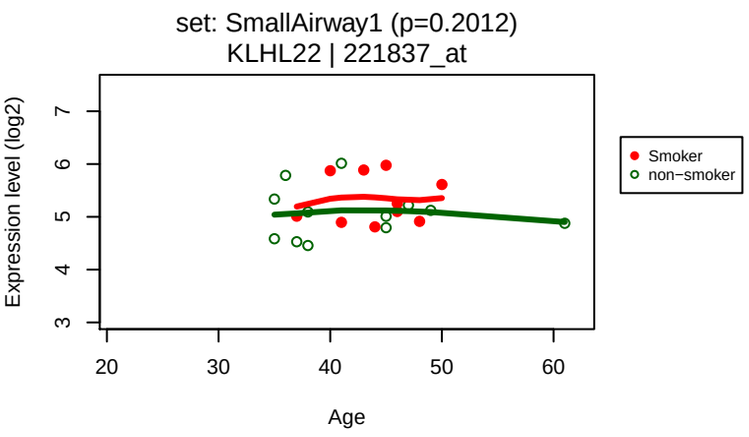
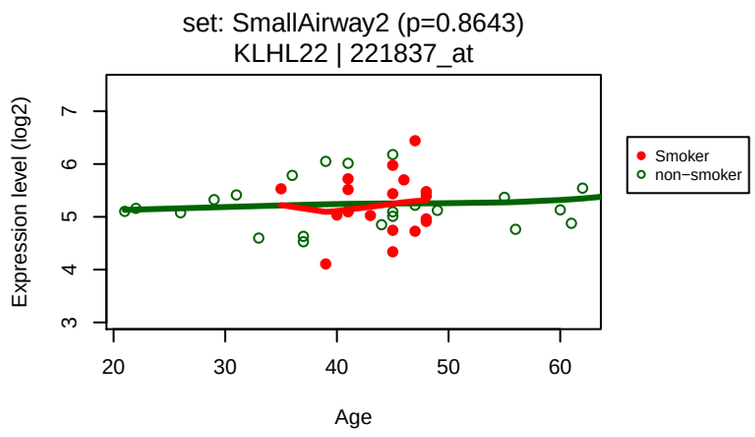
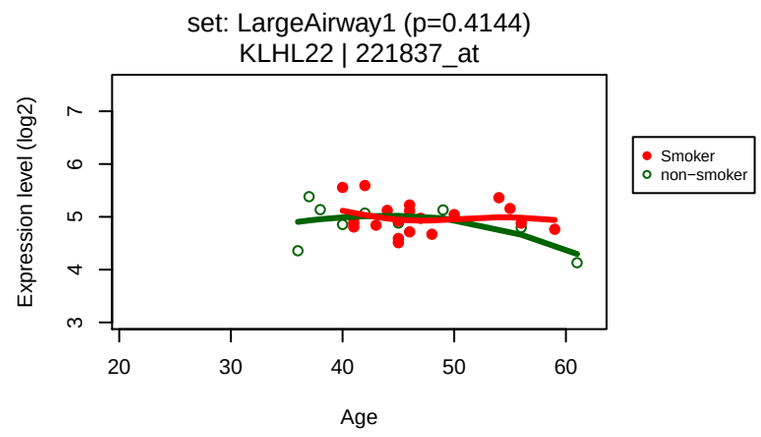

Gene expression of smokers and nonsmokers
gene: KLHL22 | 221837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2012
p=0.8643
p=0.8768
p=0.4144
p=0.5262
p=0.0913
p=0.8478
p=0.2877
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

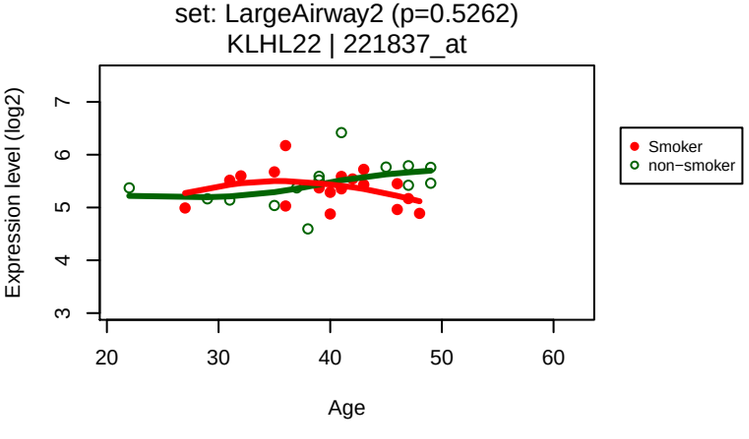

Gene expression ratio (smokers/nonsmokers)
gene: KLHL22 | 221837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2012
p=0.8643
p=0.8768
p=0.4144
p=0.5262
p=0.0913
p=0.8478
p=0.2877
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

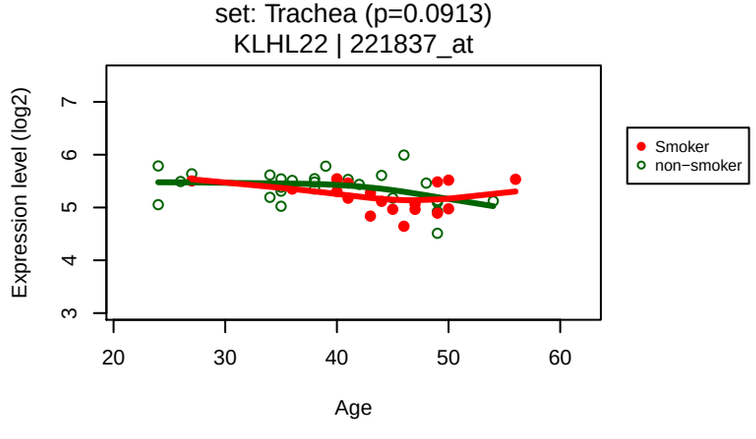

Gene expression ratio in Males
gene: KLHL22 | 221837_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.592
p=0.5445
p=0.2041
p=0.0205
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL22 | 221837_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4197
p=0.136
p=0.5168
p=0.2341
SmallAirway2
LargeAirway2
Trachea
Macrophage2

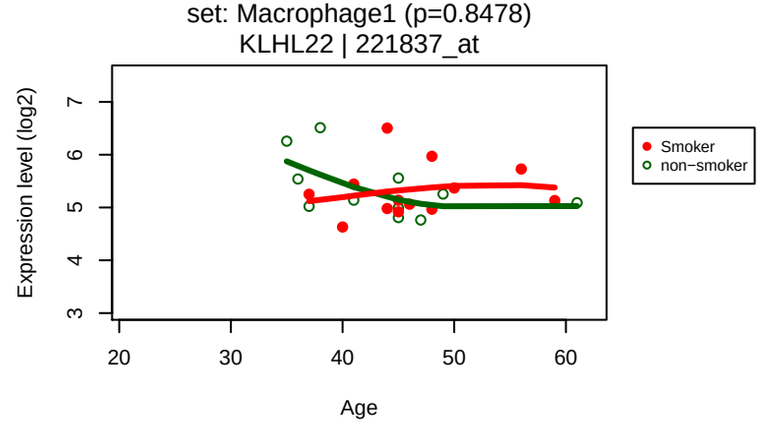

set: SmallAirway1 (p=0.2012)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker

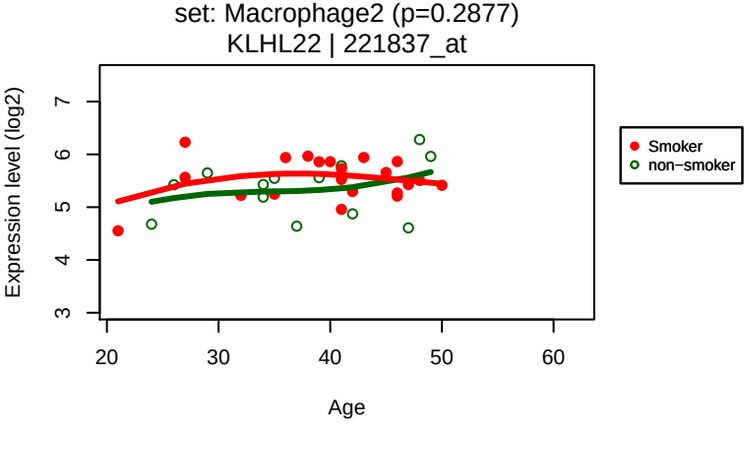

set: SmallAirway2 (p=0.8643)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker

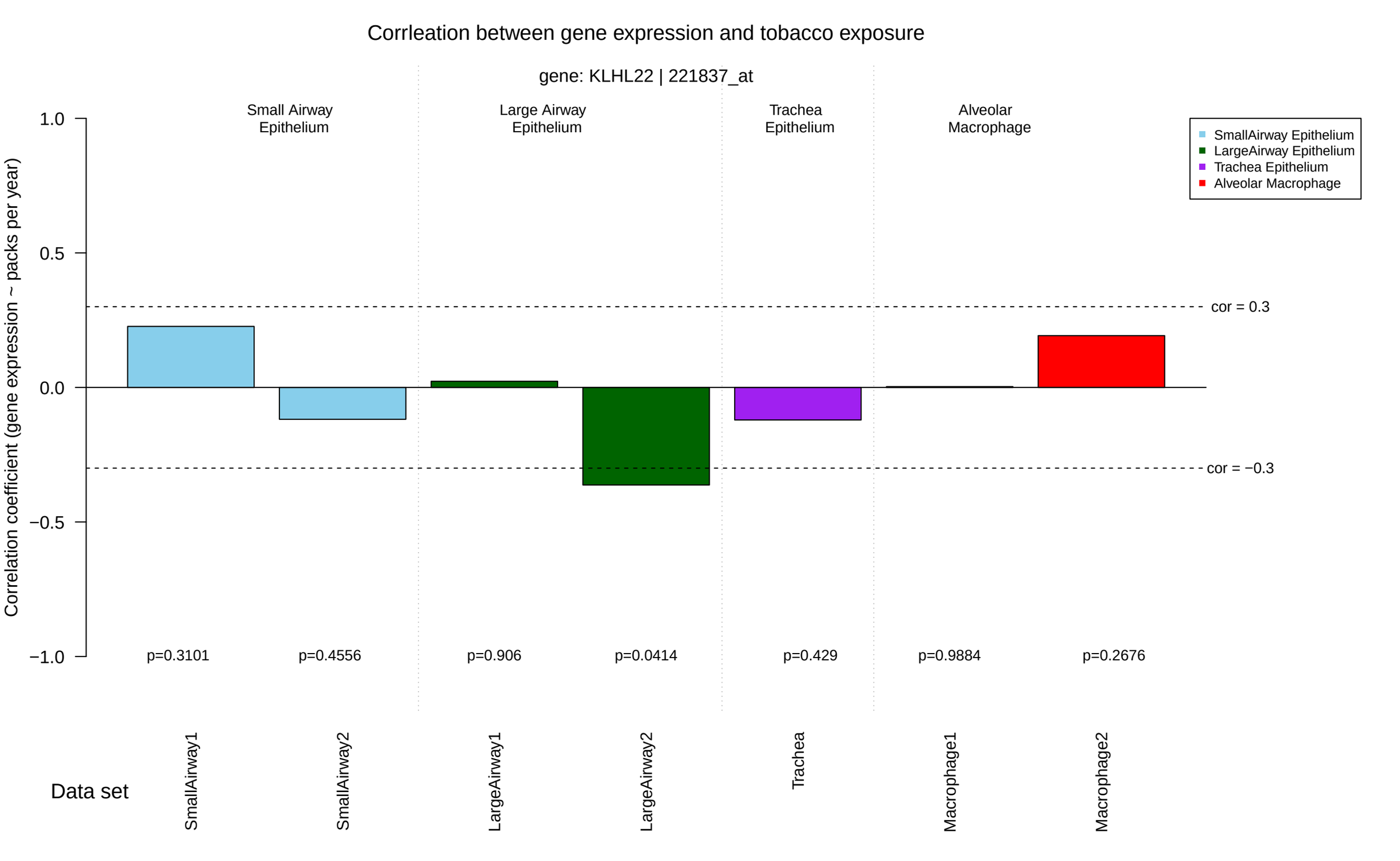

set: LargeAirway1 (p=0.4144)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5262)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0913)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8478)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2877)
KLHL22 | 221837_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL22 | 221837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3101
p=0.4556
p=0.906
p=0.0414
p=0.429
p=0.9884
p=0.2676
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2